Spivak Chapter 9 Problems

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1 Problem 5

Find f' if $f(x) = [x]$.

$f(x)$ is discontinuous at integer values of x, so the derivative does not exist there. For non-integer values of x, $f'(x) = 0$.

2 Problem 13

Suppose that $f(a) = g(a)$ and that the left-hand derivative of f at a equals the right-hand derivative of g at a. Define $h(x) = f(x)$ for $x \leq a$, and $h(x) = g(x)$ for $x \geq a$. Prove that h is differentiable at a.

Because $h(x) = f(x)$ for $x \leq a$, $h'(x) = f'(x)$. Similarly, for $x \geq a$, $h'(x) = g'(x)$. Because the left-hand derivative of $f(x)$ at a coincides with the right hand derivative of $g(x)$ at a, it follows that $h(x)$ is differentiable at a.

3 Problem 19

1. Suppose that $f(a) = g(a) = h(a)$, that $f(x) \leq g(x) \leq h(x)$ for all x, and that $f'(a) = h'(a)$. Prove that g is differentiable at a, and that that $f'(a) = g'(a) = h'(a)$.

   If $f(x) \leq g(x) \leq h(x)$ for all x, then $f'(x) \leq g'(x) \leq h'(x)$ for all x. Because $f(a) = g(a) = h(a)$ and $f'(a) = h'(a)$, $f'(a) = g'(a) = h'(a)$.

2. Show that this conclusion does not follow if we omit the hypothesis $f(a) = g(a) = h(a)$.

   Omitting the above hypothesis, it is possible to construct a $g(a)$ such that $g'(a) \neq f'(a) = h'(a)$.